

Modeling DAC Output Waveforms

Massimo D'Apuzzo, Mauro D'Arco, Annalisa Liccardo, and Michele Vadursi, *Member, IEEE*

Abstract—Digital-to-analog converters (DACs) are widely employed in control and automation, test and measurement, and communication systems. However, guidelines for the definition of static and dynamic features of DACs, as well as models capable of describing the different effects of the major nonidealities or drawbacks the DAC can suffer from, are still incomplete. This paper presents a new model capable of describing the functioning of a real DAC affected by horizontal quantization, clock modulation, vertical quantization and integral nonlinearity. The model permits evaluating the effects of the aforementioned nonidealities through straightforward formulas, which are characterized by a low computational burden. Results of experiments conducted on a real DAC to verify the reliability of the model are also shown.

Index Terms—Clock source modulation, digital-to-analog converters (DACs), horizontal quantization, integral nonlinearity (INL) errors, timing jitter, vertical quantization, zero-order-hold circuits.

I. INTRODUCTION

DIGITAL-TO-ANALOG converters (DACs) are widely employed in control and automation, test and measurement, and communication systems [1]–[3]. Electronic servo drives that control power devices, sources that generate arbitrary waveforms, and transmitters of modern communication systems that implement complex modulation schemes are just a few examples of systems that make use of DACs.

DACs operate as zero-order-hold circuits to generate a piece-constant waveform, which is aimed at approximating a smoothed waveform. To this end, the piece-constant waveform is played imposing a generation frequency much greater than the related Nyquist frequency, and low-pass filtering is performed to smooth abrupt voltage steps [4].

Modeling waveforms generated by real DACs is an open issue. In fact, models capable of describing the actual functioning of a DAC in the presence of nonidealities are still incomplete. In the past years, a theory of spectra of signals coming out of DACs has been developed. The theory explores different practical cases and defines specific approaches for each case to derive closed-form expressions for the Fourier transform of the signal [6]–[9]. All the aforementioned approaches are focused on a single drawback and neglect the effects of other coexisting

ones; furthermore, it is difficult to combine all of them in a unique general model, since they do not share a common frame [10]–[13]. Nonetheless, no models for analyzing the effects produced by a clock source affected by nonsinusoidal modulation are available in the literature [14]–[17].

The analysis of DAC output waveforms through discrete simulations is a possible alternative to the incompleteness of modeling strategies. However, simulation comes out to be impractical as there are many difficulties related to memory resources requirements, computational burden, and measurement time that need to be overcome. The most challenging hurdle is that a huge number of points is needed to adequately describe the minimal timing variations due to jitter that can affect all the constant pieces of the waveform. At the same time, trying to reduce the complexity of simulations by means of rough approximations can definitely lead to erroneous estimates [18].

Designers and technicians thus have poor support to evaluate the impact of different choices of generation frequency and types of output filter and predict the drawbacks produced by nonidealities. Furthermore, without a reliable model the uncertainty of the parameters related to the output waveform cannot be estimated [5].

This paper formulates a new model that gives a thorough insight into DAC functioning by expressing the major DAC nonidealities in the form of additive terms. The proposed model pays attention to the piece-constant nature of waveforms generated by DACs by introducing the concept of *horizontal quantization*. It also takes into account three other important mechanisms connected with the waveform generation that can be responsible for drawbacks: the possible modulation of the clock source, the limited number of bits, and the integral nonlinearity (INL).

The model is developed in both the time and the frequency domain. Thanks to some useful manipulations, the spectrum of the output waveform is expressed in the form of a discrete Fourier transform (DFT). It is thus possible to predict the behavior of a real DAC in terms of the spectral distortion of its output waveform, given the entity of the drawbacks, in a more efficient way than through computationally heavy simulations. This is the main merit of this paper. Another one is that the model takes into account all the major drawbacks of DACs and can be applied to any type of deterministic jitter and not only to the sinusoidal jitter.

This paper is organized as follows. The major DACs drawbacks are introduced in Section II. The time-domain and frequency-domain models are presented in Sections III and IV, respectively. The results of laboratory experiments carried out on a real DAC whose drawbacks are modifiable and known are described in Section V. Finally, conclusions are given in Section VI.

Manuscript received November 3, 2009; revised January 12, 2010; accepted January 13, 2010. Date of publication April 19, 2010; date of current version October 13, 2010. The Associate Editor coordinating the review process for this paper was Dr. Wendy Van Moer.

M. D'Apuzzo, M. D'Arco, and A. Liccardo are with the Dipartimento di Ingegneria Elettrica, Università degli Studi di Napoli Federico II, 80125 Napoli, Italy (e-mail: dapuzzo@unina.it; darco@unina.it; aliccard@unina.it).

M. Vadursi is with the Dipartimento per le Tecnologie, Università degli Studi di Napoli Parthenope, 80143 Napoli, Italy (e-mail: michele.vadursi@uniparthenope.it).

Digital Object Identifier 10.1109/TIM.2010.2046649

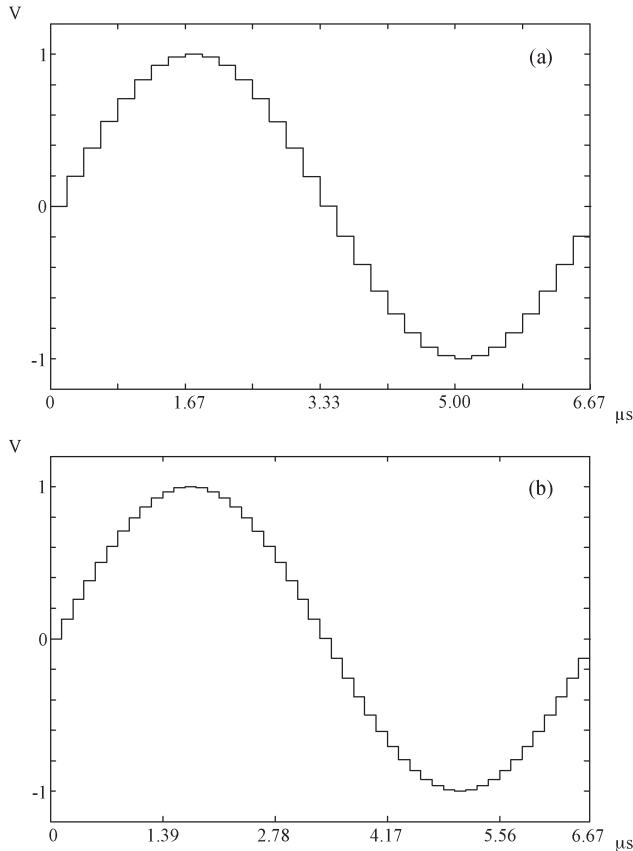


Fig. 1. Graphic sketches the DAC output of a 150-kHz sinusoid: (a) the generation frequency is 4.8 MS/s, and the number of points per period M is 32; (b) the generation frequency is 7.2 MS/s, and the number of points per period M is 48.

II. DAC DRAWBACKS

A. Horizontal Quantization

All the waveforms generated by DACs cannot hide a built-in discrete nature in time, since they are obtained by changing the output voltage with step variations at discrete time instants (Fig. 1). With respect to the target waveforms, which are smoothed, those played by means of DACs are characterized by spurious components in the spectrum, which are originated by such step variations.

Likewise, the term *vertical quantization* is commonly accepted to refer to amplitude discretization, and the term *horizontal quantization* can be adopted to refer to the discrete-time nature of DAC waveforms. Horizontal quantization becomes the main degradation factor when the generation frequency is not sufficiently greater than the Nyquist frequency of the expected waveform. In this case, spurious terms due to horizontal quantization appear closer to the bandwidth of the target waveform.

B. Clock Modulation

In addition to horizontal quantization, further spurious components appear in the spectrum of the output waveform when the clock source has poor stability. In particular, the main cause of clock inaccuracy is due to a possible superimposed deterministic modulation, which is responsible for periodic

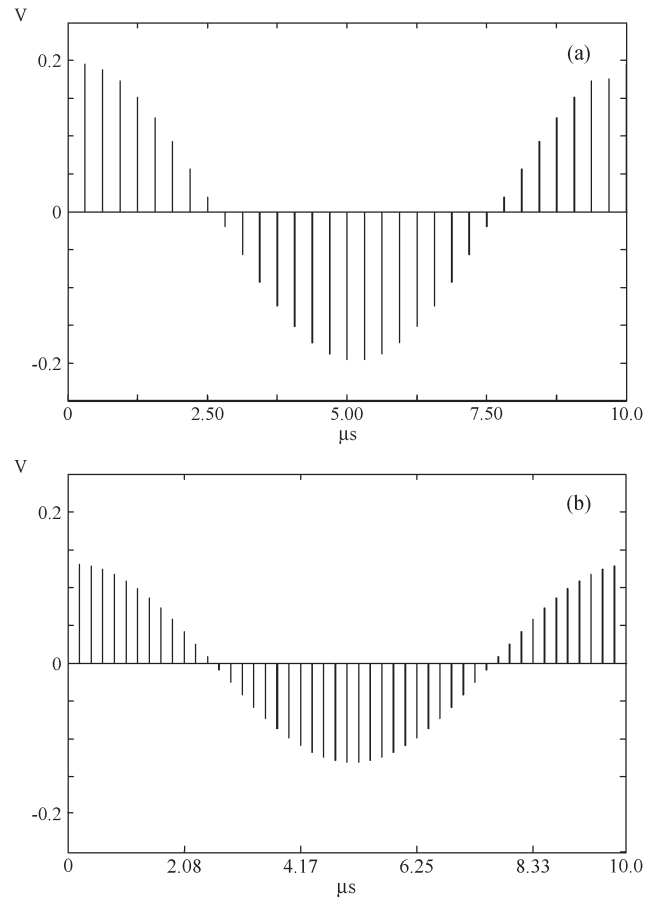


Fig. 2. Clock modulation effects consist of a discrete series of short pulses: (a) refers to Fig. 1(a), whereas (b) is related to Fig. 1(b). The two figures show that horizontal quantization magnifies the amplitude of the short pulses.

variations in the generation period. Due to the clock source instability, each voltage switching can either delay or anticipate its expected time instant. If clock modulation effects are highlighted in terms of the difference between the two piece-constant waveforms (the ideal one that is unaffected by the clock modulation and the real one), diagrams made up of a discrete series of short pulses are obtained. An analysis of a set of these diagrams, which are obtained in different conditions of clock modulation and horizontal quantization features, shows that severe horizontal quantization magnifies the amplitude of the aforementioned short pulses (Fig. 2).

C. Vertical Quantization

Any DAC is characterized by a finite voltage resolution, which is expressed in terms of the number of bits. Sample points describing a discrete waveform must therefore be approximated with the nearest voltage level that the DAC can produce in output. It is worth noting that the uniform amplitude quantization, which is necessary for DAC functioning, is the result of mathematical operations and is a deterministic effect. This is different from what occurs in the analog-to-digital conversion, where, due to several uncontrolled causes such as background noise, mismatches, and hysteresis, vertical quantization assumes random features and can be dealt with as a stochastic process.

In the presence of INL, the output values produced by a real DAC differ from that expected by an ideal one. In such a case, the vertical quantization is nonuniform. To take into account INL effects, the actual transfer function of the DAC, which provides the actual output values that the DAC produces, should be known.

Vertical quantization effects are weakly coupled with both horizontal quantization and poor time base stability: by evaluating the difference between vertical quantization terms, which are obtained in the presence of different combinations of horizontal quantization and clock modulation drawbacks, no relevant amplitude variations are recognized.

III. MODELING THE DAC OUTPUT SIGNAL IN THE TIME DOMAIN

Let $x(t)$ be the target waveform to be generated by a DAC. The waveform is defined by a discrete sequence $\{x(n)\}$ consisting of sample points of $x(t)$ uniformly spaced in time. The piece-constant waveform $y_x(t)$, which is produced using a constant generation period T_g , differs from $x(t)$, because of the presence of the three main drawbacks described in the previous section. The proposed model accounts for the drawbacks effects as three additive terms: $y_x^{\text{HQ}}(t)$ due to horizontal quantization, $y_x^{\text{CM}}(t)$ due to clock source modulation, and $y_x^{\text{VQ}}(t)$ due to vertical quantization, with the latter including a possible INL. Thus, the DAC output waveform can be written as

$$y_x(t) = x(t) + y_x^{\text{HQ}}(t) + y_x^{\text{CM}}(t) + y_x^{\text{VQ}}(t). \quad (1)$$

Horizontal quantization effects can be expressed by

$$y_x^{\text{HQ}}(t) = \sum_{n=-\infty}^{\infty} x(n)g\left(\frac{t - nT_g}{T_g}\right) - x(t) \quad (2)$$

in which $g(\theta)$ represents a unitary pulse function with unitary duration in the normalized time variable θ , i.e.,

$$g(\theta) = \begin{cases} 1, & 0 \leq \theta < 1 \\ 0, & \text{elsewhere.} \end{cases} \quad (3)$$

Concerning clock modulation effects, it is worth noting that voltage switchings can either delay or anticipate the expected time instant nT_g by a time amount Δ_n , which is positive in the case of delay and negative otherwise. The effects of clock modulation can be modeled by an additive term that contains a discrete sequence of short pulses typically disjoint from each other, i.e.,

$$y_x^{\text{CM}}(t) = \sum_{n=-\infty}^{\infty} x(n)h_n(t - nT_g) \quad (4)$$

where

$$h_n(t) = -\text{sign}(\Delta_n)g\left(\frac{t - nT_g}{\Delta_n}\right) + \text{sign}(\Delta_{n+1})g\left(\frac{t - (n+1)T_g}{\Delta_{n+1}}\right). \quad (5)$$

The function $h_n(t)$ is expressed in terms of signum functions $\text{sign}(\cdot)$ and a pair of consecutive pulses that are scaled and folded versions of $g(\theta)$. In detail, if Δ_n is positive (i.e., if it represents a delay), the rectangular pulse $g(t/\Delta_n)$ exhibits a rising edge at time instant zero and a falling edge at time instant Δ_n ; else, if Δ_n is negative (i.e., it represents a lead), the rising edge is at the negative time instant Δ_n , and the falling edge is at zero. The duration of the generic pulse is given by the absolute time amount $|\Delta_n|$ of the delay or lead; the polarity of the pulse depends on the signum of Δ_n .

Vertical quantization implies that the sample points describing the discrete waveform $x(n)$ are approximated by the nearest voltage level that the DAC can produce in output, namely, $\hat{x}(n) + T[\hat{x}(n)]$, where $T[\hat{x}(n)]$ is the INL of the DAC for the quantized voltage level $\hat{x}(n)$. Vertical quantization effects can consequently be described by the following expression:

$$y_x^{\text{VQ}}(t) = \sum_{n=-\infty}^{\infty} \{\hat{x}(n) + T[\hat{x}(n)] - x(n)\} \times \left[g\left(\frac{t - nT_g}{T_g}\right) + h_n(t - nT_g) \right]. \quad (6)$$

It is worth noting that vertical quantization is uniform for ideal DACs, which do not suffer from INL (i.e., $T[\hat{x}(n)] = 0$ for any n), whereas the INL is responsible for a nonuniform vertical quantization.

IV. MODELING DAC OUTPUT SIGNAL IN THE FREQUENCY DOMAIN

The analysis of the terms in (2), (4), and (6) is conducted here with reference to the case of continuous-generation mode of a signal $x(t)$, which is characterized by a limited bandwidth B and represented by M sample points $\{x(n)\}_{n=1,\dots,M}$. In continuous-generation mode, the DAC replicates the finite-length sequence made up of M samples to produce a periodic signal characterized by a period $T_0 = MT_g$, with T_g being the generation period of the DAC; the reciprocal of T_0 and T_g represent, respectively, the repetition rate f_0 and the generation rate f_g . Continuous-generation mode is the standard mode adopted to grant steady-state test conditions.

When the DAC is affected by a periodic clock modulation, it is easy to show that the additive terms $y_x^{\text{CM}}(t)$, $y_x^{\text{VQ}}(t)$, and $y_x^{\text{NL}}(t)$ are periodic as well, and their period T_Ω is the least common multiple of T_0 , T_g , and T_m , the last one being the clock modulation period (i.e., the reciprocal of the modulation fundamental frequency f_m). In other words, there are three positive integers, i.e., U , V , and Z ($Z = MU$), among which U and V are prime to each other, that satisfy the following equation:

$$UT_0 = VT_m = ZT_g = T_\Omega \quad (7)$$

or, equivalently

$$\frac{f_0}{U} = \frac{f_m}{V} = \frac{f_g}{Z} = f_\Omega. \quad (8)$$

Note that, in the presence of drawbacks, the actual period of the output signal $y_x(t)$ is T_Ω ; its reciprocal f_Ω , which is the maximum common divisor between f_0 , f_g , and f_m , is the fundamental frequency of the output.

The values of the parameters f_0 , f_g , and f_m have been considered commensurable, as it is usual in practical approaches. In fact, all the parameters are observed and measured within a finite time interval, which implies a finite frequency resolution.

As each of the three addends that describe the nonidealities of the DAC at the right-side term of (1), i.e., $y^{\text{HQ}}(t)$, $y^{\text{CM}}(t)$, and $y^{\text{VQ}}(t)$, is periodic, the spectral analysis is conducted in the following by representing them in terms of their Fourier series expansion. The scope is to estimate the coefficients of the series from the samples of the signal through a DFT to have a straightforward insight in the frequency domain. The Fourier series expansions will have fundamental frequency f_Ω , which is common to all the additive terms.

A. Horizontal Quantization

The Fourier series expression of $y^{\text{HQ}}(t)$ is

$$y_x^{\text{HQ}}(t) = \sum_{k=-\infty}^{\infty} c_k^{\text{HQ}} e^{j2\pi k f_\Omega t}. \quad (9)$$

The coefficients c_k^{HQ} are defined as

$$c_k^{\text{HQ}} = \frac{1}{T_\Omega} \int_0^{T_\Omega} y_x^{\text{HQ}}(t) e^{-j2\pi k f_\Omega t} dt \quad (10)$$

and can be expressed by

$$c_k^{\text{HQ}} = \frac{1}{Z} \text{sinc}\left(\frac{k}{Z}\right) e^{-j\frac{\pi}{2}k} \sum_{n=0}^{Z-1} x(n) e^{-j\frac{2\pi}{Z}kn} - \frac{1}{T_\Omega} X_L(k f_\Omega) \quad (11)$$

(see Appendix A for details). In (11)

$$\sum_{n=0}^{Z-1} x(n) e^{-j\frac{2\pi}{Z}kn} = \text{DFT}_Z[x(n)](k), \quad 0 \leq k < Z. \quad (12)$$

Namely, for k that ranges from 0 to $Z - 1$, the term expresses the DFT of the sequence $x(n)$ calculated on Z points. Letting k range from $-\infty$ to $+\infty$, the periodic repetition, at any Z points, of $\text{DFT}_Z[x(n)](k)$ values, is obtained. $X_L(\cdot)$ is a version of the spectrum of the signal $x(t)$ affected by leakage.

Note that, for frequencies $k f_\Omega$ that are outside the bandwidth of the waveform $x(t)$, i.e., for $|k| > Z/2$, $X_L(k f_\Omega)$ is zero, whereas it almost counterbalances the first term in (11) for $|k| < Z/2$. Horizontal quantization effects, which are described by a rapidly varying term $y^{\text{HQ}}(t)$, are mostly to be found outside the bandwidth of $x(t)$, where

$$c_k^{\text{HQ}} \approx \frac{1}{Z} \text{sinc}\left(\frac{k}{Z}\right) e^{-j\pi \frac{k}{Z}} \text{DFT}_Z[x(n)](k), \quad |k| > \frac{Z}{2}. \quad (13)$$

A thorough analysis of (13) shows that the main effect of horizontal quantization is the growth of a discrete number of

attenuated replicas of the spectrum of $x(t)$, which are spaced in the frequency domain at a pace equal to the generation rate f_g .

B. Clock Modulation

The complex coefficients c_k^{CM} of the Fourier series of the term $y_x^{\text{CM}}(t)$ are given by

$$c_k^{\text{CM}} = \frac{1}{T_\Omega} \int_0^{T_\Omega} y_x^{\text{CM}}(t) e^{-j2\pi k f_\Omega t} dt. \quad (14)$$

Commonly, clock modulation effects are very weak. The only ones that need be taken into consideration are those that produce spectral growth within the bandwidth B . The attention is therefore focused on the sole coefficients related to the values of k that range from $-Z/2$ to $Z/2$, which are largely sufficient to describe the amplitude spectrum of $y^{\text{CM}}(t)$. Such coefficients can be expressed as (see Appendix B for details)

$$c_k^{\text{CM}} = -\frac{1}{T_\Omega} \sum_{n=1}^{Z-1} [x(n) - x(n-1)] \Delta_n e^{-j\frac{2\pi}{Z}kn}. \quad (15)$$

Note that the sequence $\{FD[x(n)] = x(n) - x(n-1)\}$ represents the first finite difference of the digital waveform $x(n)$. This is a useful result to evaluate the effects of clock modulation on piece-constant waveforms, because it proves that the contributions to the spectrum of the output signal, which are given by the c_k^{CM} , values can easily be estimated by applying the algorithm DFT_Z to $FD[x(n)]$ windowed by $\{\Delta_n\}$, i.e.,

$$c_k^{\text{CM}} = -\frac{1}{T_\Omega} \text{DFT}_Z\{FD[x(n)] \Delta_n\}. \quad (16)$$

It is worth highlighting that the algorithm DFT_Z allows transforming the product of two digital sequences in the convolution of the two separately transformed sequences (convolution property). For example, if $FD[x(n)]$ are the samples of a sinusoid at frequency f_0 and Δ_n is sinusoidal as well at frequency f_m , the convolution property straightforwardly shows that the only nonzero values of c_k^{CM} within the frequency interval $(0, B)$ are found at frequencies $|f_0 \pm f_m|$, and their amplitude is equal to half of the product of the amplitudes of both sinusoids.

C. Vertical Quantization

The complex coefficients c_k^{VQ} of the Fourier series of the term $y_x^{\text{VQ}}(t)$ are given by

$$c_k = \frac{1}{T_\Omega} \int_0^{T_\Omega} \sum_{n=0}^{Z-1} \left[(T[\hat{x}(n)] + \hat{x}(n) - x(n)) \times \left(g\left(\frac{t - nT_g}{T_g}\right) + h_n(t - nT_g) \right) \right] e^{-j2\pi k f_\Omega t} dt. \quad (17)$$

Thanks to passages very similar to those made for the terms $y_x^{\text{CM}}(t)$ and $y_x^{\text{HQ}}(t)$, the Fourier series coefficients can

straightforwardly be written as

$$c_k^{\text{VQ}} = \frac{1}{Z} \text{sinc} \left(\frac{k}{Z} \right) \times e^{-j \frac{\pi}{Z} k} \sum_{n=0}^{Z-1} (T[\hat{x}(n)] + \hat{x}(n) - x(n)) e^{-j \frac{2\pi}{Z} kn} + \frac{1}{T_\Omega} \sum_{n=0}^{Z-1} FD[T[\hat{x}(n)] + \hat{x}(n) - x(n)] \Delta_n e^{-j \frac{2\pi}{Z} kn}. \quad (18)$$

Again, the coefficients can be estimated through a DFT_Z, namely as

$$c_k^{\text{VQ}} = \frac{1}{Z} \text{sinc} \left(\frac{k}{Z} \right) e^{-j \frac{\pi}{Z} k} \text{DFT}_Z [T[\hat{x}(n)] + \hat{x}(n) - x(n)] + \frac{1}{T_\Omega} \text{DFT}_Z \{ FD[T[\hat{x}(n)] + \hat{x}(n) - x(n)] \Delta_n \}. \quad (19)$$

V. EXPERIMENTAL ANALYSIS

An experimental analysis of the waveform at the output of a DAC is carried out in the frequency domain. The results are then compared to those expected from the model to evaluate the model's reliability. To design meaningful experiments, two main challenges have to be faced: the DAC's drawbacks have to be known and under control, and the spectral analysis has to be accurate enough. To meet such goals, a suitable measurement testbed is set up. It consists of 1) a control and processing unit (PC); 2) a binary-offset 12-bit DAC, which is inserted in the architecture of an arbitrary waveform generator Sony-Tektronix AWG2020 (250-MS/s maximum generation frequency); 3) a function generator (Agilent Technologies 33120A); and 4) a spectrum analyzer (Hewlett Packard 8594E). All the instruments are interconnected through a standard IEEE-488 interface bus.

The PC synthesizes the digital version $x(n)$ of the test waveform to be downloaded into the memory of the arbitrary waveform generator and played by means of its internal DAC in analog form. The arbitrary waveform generator operates with an external clock, which is provided by the function generator. Thus, sinusoidal jitter can be introduced by imposing an FM to the generator's output signal. Furthermore, the output low-pass filter of the arbitrary waveform generator is bypassed to avoid the strong attenuation that would cancel out the out-of-band effects of some drawbacks, particularly those due to horizontal quantization.

The spectrum analyzer gives the power spectrum of the signal generated by the DAC. The use of an analog spectrum analyzer rather than a digital fast Fourier transform (FFT) analyzer is necessary here to assure sufficient sensitivity (i.e., low noise floor) to highlight the weak effects produced by DAC drawbacks. In fact, FFT analyzers that perform analog-to-digital conversion could introduce themselves vertical quantization and INL effects, which would hardly be distinguished by those related to the DAC under test.

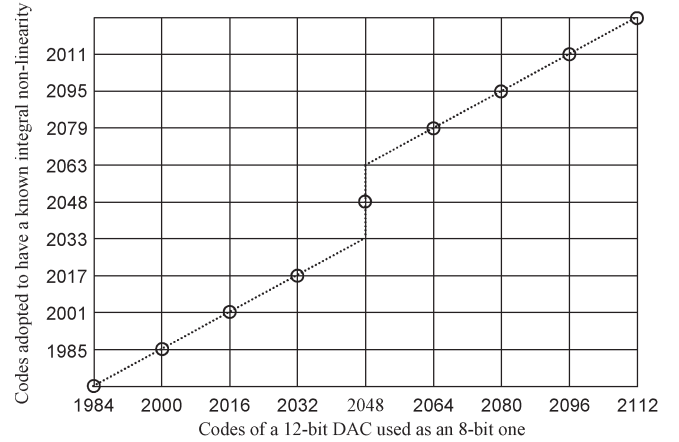


Fig. 3. Graphic describes the INL considered in the experiment.

Hereafter, the results of a single test are shown and compared to those expected from the proposed model. The waveform settings are chosen to have understandable issues to draw commentary. Therefore, the presented results are not intended to be exhaustive at all, since the model is very general, not limited to specific types of waveforms, and suitable to analyze any type of periodic jitter.

The test is related to a sine wave synthesized with 32 points per period and played using an average clock running at a frequency f_g equal to 4.8 MHz (the sine wave is thus characterized by a nominal frequency f_0 equal to 150 kHz). The clock is jittered imposing an FM sinusoidal modulation, which is characterized by a frequency f_m equal to 20 kHz and a peak frequency deviation Δ_P equal to 0.1% (i.e., 4.8 kHz). According to these settings, $f_\Omega = 10$ kHz, $U = 15$, $V = 2$, and $Z = 480$.

For synthesizing the waveform, a set of 255 different amplitude values is considered, thus producing the uniform vertical quantization effects related to an 8-bit DAC. This choice, which might seem unnatural for a 12-bit DAC since it involves forcing to zero the four less significant bits of the converter, allows introducing a known nonuniform vertical quantization, i.e., an INL. The four less significant bits, which are not actually forced to zero, are exploited to impose an exemplificative nonlinearity, which is responsible for a discontinuity in the DAC transfer function between negative and positive voltage values, as sketched in Fig. 3. In fact, the DAC used in the experiments is a 12-bit binary-offset converter, with codes ranging from 0 to 4095. Utilizing it as an 8-bit DAC means that only codes that are multiples of 16 should be considered. The use of a neighboring code, in place of that belonging to the set of multiples of 16, allows the insertion of a known nonlinearity. The nonlinearity inserted for test purposes produces the discontinuity in the DAC transfer function, which is shown in Fig. 3.

In detail, Fig. 4 shows the spectrum of the waveform produced by the DAC under test as it is measured by the spectrum analyzer that spans the range of 50 kHz–20 MHz. The test waveform is characterized by a main spectral line at 150 kHz; however, this is not the sole spectral contribution because of the presence of several drawbacks. The spectral lines located at frequency values $f_g \pm f_0 = (4.80 \pm 0.15)$ MHz,

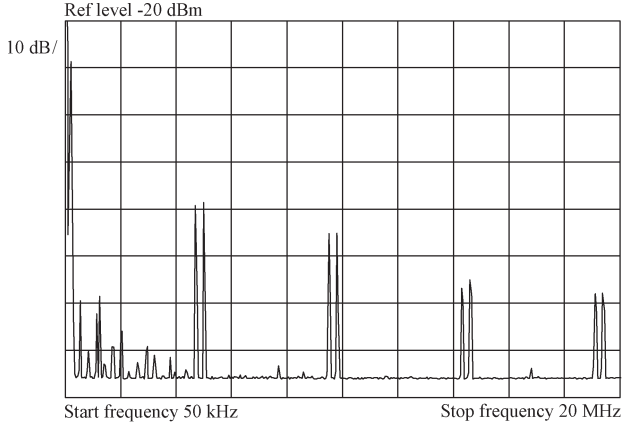


Fig. 4. Spectrum of the waveform produced by the DAC under test.

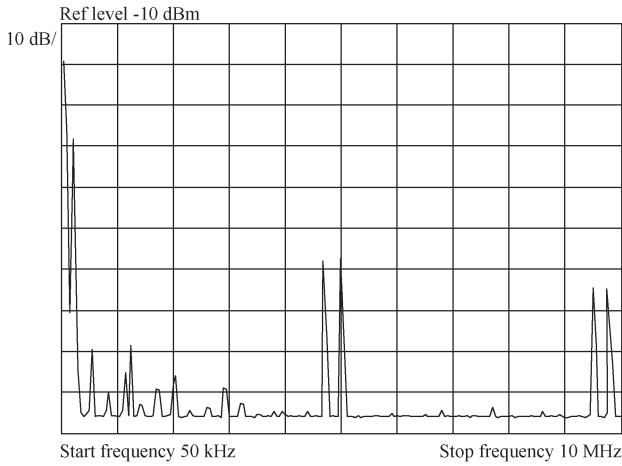


Fig. 5. Results related to the analysis of horizontal quantization effects.

$2f_g \pm f_0 = (9.60 \pm 0.15)$ MHz, $3f_g \pm f_0 = (14.40 \pm 0.15)$ MHz, and $4f_g \pm f_0 = (19.20 \pm 0.15)$ MHz are due to horizontal quantization, whereas those closer to the main spectral line are due to the other drawbacks; some spurious contributions are also present. From Fig. 4, it is difficult to draw accurate estimations, since the large frequency span (almost 20 MHz) does not grant a frequency resolution that is fine enough to distinguish all the contributions. Furthermore, the choice of a reference level equal to -20 dBm, which involves a soft attenuation of the waveform under analysis, makes distortion terms appear as well. These terms, which are typically introduced by the mixer inside the analog spectrum analyzer when it has to work in the presence of waveform containing more spectral contributions, could erroneously be regarded as DAC defects. Furthermore, the considered settings for the spectrum analyzer are not suitable to avoid the local oscillator feedthrough, which is responsible of the fictitious spectral contents that appears at the very low frequencies of the spanned range in Fig. 4.

To draw meaningful comparisons between the results offered by the proposed model and the experimental ones, the effects of each drawback is analyzed by selecting specific and more adequate settings for the spectral analysis. In particular, Fig. 5 shows the results related to the analysis of the horizontal quantization effects. A reference level equal to 0 dBm and a frequency

span that ranges from 50 kHz to 10 MHz allow more accurate measurements of the power of the spectral lines residing at frequencies of 4.65 , 4.95 , 9.45 , and 9.75 MHz. According to the proposed model, the value of these contributions can be estimated as

$$2|c_k| = 2 \left| \frac{1}{Z} \text{sinc} \left(\frac{k}{Z} \right) e^{-j\pi \frac{k}{Z}} \text{DFT}_Z [x(n)](k) \right| \quad (20)$$

for k equal to $Z - U = 465$, $Z + U = 495$, and $2Z \pm U$. In relative terms, the spectral lines due to horizontal quantization are expected to be $20 \log |\text{sinc}(k/Z)|$ dB below the main spectral line at 150 kHz, which is characterized by a power level equal to -28 dBm. The values expected from the proposed model and those measured through the spectrum analyzer are given in Table I. The difference between the expected and measured values is below the measurement uncertainty of the spectrum analyzer (typically ± 2 dB).

Fig. 6 shows the features of the spectrum very near the main component at 150 kHz and highlights the contributions due to clock modulation, which consist of weak spectral lines at both the left and right sides of the main one. To estimate the amplitude of these weak components, the sequence Δ_n that describes the accumulated jitter has to be processed. A simple expression for the sequence Δ_n can be achieved from the nominal data related to clock modulation. Note that the initial phase of the periodic accumulated jitter is a random variable. It does not influence the value of the amplitude of the spectral line, i.e., a meaningful estimation can be obtained even ignoring the phase information. In particular, in the case of sinusoidal modulation characterized by the given values of f_m and Λ_P , the following expression can be used to describe Δ_n and carry out the estimation of clock modulation effects (see Appendix C for details):

$$\Delta_n = \frac{\Lambda_P}{\pi f_m} \sin \left(\pi \frac{V}{Z} \right) \cos \left(2\pi \frac{V}{Z} n \right). \quad (21)$$

Taking into account that the finite difference of the digital waveform is sinusoidal with amplitude equal to $2\pi U/Z$ times that of the input waveform, the spectral lines due to jitter effects, which are located at frequencies of 130 and 170 kHz, which are obtained for $k = U - V = 13$ and $k = U + V = 17$, respectively, have to be

$$20 \log \left| 2\pi \frac{U \Lambda_P}{Z} \right| \text{ dB} \quad (22)$$

below the reference level at 0 dBm. The results are summarized in Table II.

Fig. 7 points out the effects of nonuniform vertical quantization. An integral nonlinearity consisting, in this simple case, of odd harmonics has been introduced. From Fig. 7, the third and fifth harmonics can be measured and compared to the estimation performed through (19). In particular, with the sequence $T[x(n)]$ in this experiment being a square wave with amplitude $15/2^{11}$ times that of the waveform under test and the

TABLE I
COMPARISON BETWEEN THE EXPECTED AND EXPERIMENTAL RESULTS FOR THE HORIZONTAL QUANTIZATION

Frequency	4.65 MHz	4.95 MHz	9.45 MHz	9.75 MHz
Expected Power	-58 dBm	-58 dBm	-64 dBm	-64 dBm
Measured Power	-59 dBm	-59 dBm	-65 dBm	-65 dBm

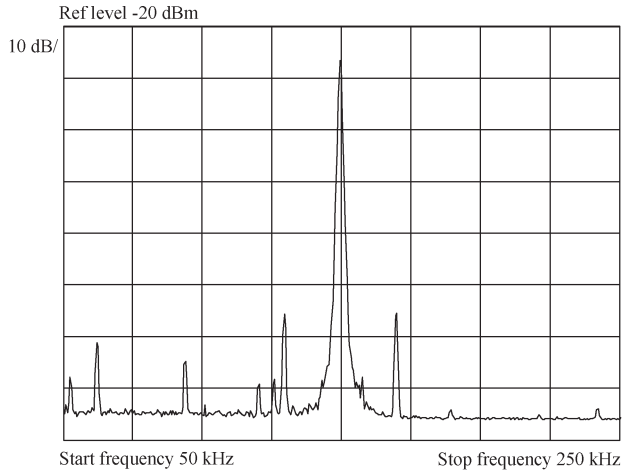


Fig. 6. Results related to the analysis of clock modulation.

TABLE II
COMPARISON BETWEEN THE EXPECTED AND EXPERIMENTAL RESULTS FOR THE CLOCK MODULATION

Frequency	130 kHz	170 kHz
Expected Power	-74 dBm	-74 dBm
Measured Power	-75 dBm	-75 dBm

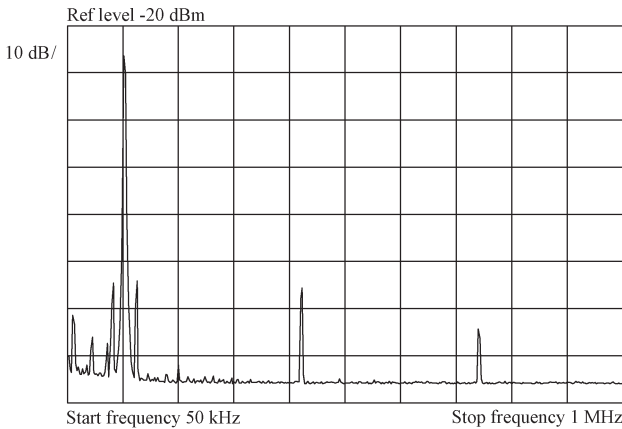


Fig. 7. Results related to the analysis of the nonuniform vertical quantization due to the INL shown in Fig. 3.

first term in (19) being dominant over the second one, the 3rd and 5th harmonics are expected to be

$$20 \log \left| \frac{4}{\pi m} \text{sinc} \left(\frac{mU}{Z} \right) \frac{15}{2^{11}} \right| \quad (23)$$

below the fundamental spectral component (m is equal to, respectively, 3 and 5). Results of estimations are given in Table III

TABLE III
COMPARISON BETWEEN THE EXPECTED AND EXPERIMENTAL RESULTS FOR THE NONUNIFORM VERTICAL QUANTIZATION

Frequency	450 kHz	750 kHz
Expected Power	-78 dBm	-83 dBm
Measured Power	-77 dBm	-84 dBm

for the spectral lines at frequencies of 450 and 750 kHz, which are obtained for $k = 3U = 45$ and $k = 5U = 75$, respectively.

VI. CONCLUSION

A new model for the analysis of waveforms generated by DACs has been presented. The model includes analytical expressions, in both the time and frequency domains, for describing the main drawbacks that affect the DAC functioning. All the analytical expressions have adequately been processed to derive straightforward formulas, which can be evaluated through fast and well-known algorithms. The proposed model represents a viable solution to predict the effects of typical drawbacks, overcoming the limitations of discrete simulated approaches that often fail their intent because of a heavy computational burden and excessive memory resource requirements. A suitable measurement testbed has been designed and set up to experimentally analyze the reliability of the model. The difference between the power values expected from the model and those measured is below the measurement uncertainty of the spectrum analyzer.

Ongoing research is aimed at expanding the model to include other DAC nonidealities such as glitches and transients and exploiting the model to design and calibrate a reference generator.

APPENDIX A

$$\begin{aligned} c_k^{\text{HQ}} &= \frac{1}{T_\Omega} \int_0^{T_\Omega} y_x^{\text{HQ}}(t) e^{-j2\pi k f_\Omega t} dt \\ &= \frac{1}{T_\Omega} \left[\int_0^{T_\Omega} \sum_{n=-\infty}^{\infty} x(n) g \left(\frac{t - nT_g}{T_g} \right) e^{-j2\pi k f_\Omega t} dt \right. \\ &\quad \left. - \int_0^{T_\Omega} x(t) e^{-j2\pi k f_\Omega t} dt \right]. \end{aligned} \quad (\text{A.1})$$

By observing that, in the interval $(0, T_g)$, the index n has to range from 0 to $Z - 1$ and introducing the window function

$W_{[0,T_\Omega]}$, which is unitary in $[0, T_\Omega]$ and zero elsewhere, the extremes of the integration of both integrals can be extended to $(-\infty, +\infty)$, yielding

$$c_k^{\text{HQ}} = \frac{1}{T_\Omega} \left[\sum_{n=0}^{Z-1} x(n) \int_{-\infty}^{\infty} g\left(\frac{t-nT_g}{T_g}\right) e^{-j2\pi k f_\Omega t} dt - \int_{-\infty}^{\infty} x(t) W_{[0,T_\Omega]} e^{-j2\pi k f_\Omega t} dt \right]. \quad (\text{A.2})$$

It is easy to show that the second term in the right side of (A.2) is the sample of the spectrum of the signal $x(t)$ affected by leakage, which is taken at $k f_\Omega$. In fact

$$\int_{-\infty}^{\infty} x(t) W_{[0,T_\Omega]} e^{-j2\pi k f_\Omega t} dt = \{\Im[x(t)] \otimes \Im[W_{[0,T_\Omega]}]\} (k f_\Omega) = X_L(k f_\Omega) \quad (\text{A.3})$$

where $\Im[x(t)]$ is the Fourier transform of $x(t)$, $\otimes$ denotes the convolution product, and

$$\Im[\chi_{[0,T_\Omega]}](f) = T_\Omega \text{sinc}(f T_\Omega) e^{-j\pi f T_\Omega}. \quad (\text{A.4})$$

Thus, it holds that

$$\begin{aligned} c_k^{\text{HQ}} &= \frac{T_g}{T_\Omega} \sin c(k f_\Omega T_g) e^{-j\pi k f_\Omega T_g} \sum_{n=0}^{Z-1} x(n) e^{-j2\pi k f_\Omega n T_g} \\ &\quad - \frac{1}{T_\Omega} X_L(k f_\Omega) \\ &= \frac{1}{Z} \text{sinc}\left(\frac{k}{Z}\right) e^{-j\frac{\pi}{Z} k} \sum_{n=0}^{Z-1} x(n) e^{-j\frac{2\pi}{Z} k n} \\ &\quad - \frac{1}{T_\Omega} X_L(k f_\Omega). \end{aligned} \quad (\text{A.5})$$

APPENDIX B

$$\begin{aligned} c_k^{\text{CM}} &= \frac{1}{T_\Omega} \int_0^{T_\Omega} y_x^{\text{CM}}(t) e^{-j2\pi k f_\Omega t} dt \\ &= \frac{1}{T_\Omega} \int_0^{T_\Omega} \left[\sum_{n=0}^{Z-1} x(n) h_n(t - nT_g) \right] e^{-j2\pi k f_\Omega t} dt \\ &= \frac{1}{T_\Omega} \sum_{n=0}^{Z-1} x(n) \int_0^{T_\Omega} h_n(t - nT_g) e^{-j2\pi k f_\Omega t} dt. \end{aligned} \quad (\text{B.1})$$

In the previous equation, it has been assumed that $\Delta_0 > 0$ (note that $\Delta_0 = \Delta_Z$), which has allowed limiting n to $Z - 1$.

After solving the integral

$$\begin{aligned} &\int_0^{T_\Omega} h_n(t - nT_g) e^{-j2\pi k f_\Omega t} dt \\ &= \int_0^{T_\Omega} \left[-\text{sign}(\Delta_n) g\left(\frac{t-nT_g}{\Delta_n}\right) + \text{sign}(\Delta_{n+1}) g\left(\frac{t-(n+1)T_g}{\Delta_{n+1}}\right) \right] e^{-j2\pi k f_\Omega t} dt \\ &= -\text{sign}(\Delta_n) |\Delta_n| \text{sinc}(k f_\Omega \Delta_n) e^{-j\pi k f_\Omega \Delta_n} e^{-j\pi k f_\Omega n T_g} \\ &\quad + \text{sign}(\Delta_{n+1}) |\Delta_{n+1}| \text{sinc}(k f_\Omega \Delta_{n+1}) \\ &\quad \times e^{-j\pi k f_\Omega \Delta_{n+1}} e^{-j2\pi k f_\Omega (n+1) T_g} \\ &= -\Delta_n \text{sinc}(k f_\Omega \Delta_n) e^{-j\pi k f_\Omega \Delta_n} e^{-j2\pi k f_\Omega n T_g} \\ &\quad + \Delta_{n+1} \text{sinc}(k f_\Omega \Delta_{n+1}) e^{-j\pi k f_\Omega \Delta_{n+1}} e^{-j2\pi k f_\Omega (n+1) T_g} \end{aligned} \quad (\text{B.2})$$

we have

$$\begin{aligned} c_k^{\text{CM}} &= -\frac{1}{T_\Omega} x(0) \Delta_0 \text{sinc}(k f_\Omega \Delta_0) e^{-j\pi k f_\Omega \Delta_0} + \\ &\quad - \frac{1}{T_\Omega} \sum_{n=1}^{Z-1} [x(n) - x(n-1)] \\ &\quad \times \Delta_n \text{sinc}(k f_\Omega \Delta_n) e^{-j\pi k f_\Omega \Delta_n} e^{-j2\pi k f_\Omega n T_g} \\ &\quad + \frac{1}{T_\Omega} x(Z) \Delta_Z \text{sinc}(k f_\Omega \Delta_Z) e^{-j\pi k f_\Omega \Delta_Z} e^{-j2\pi k f_\Omega Z T_g}. \end{aligned} \quad (\text{B.3})$$

With $x(0) = x(Z)$, $\Delta_Z = \Delta_0$, and $f_\Omega T_g = Z$, the first and last terms cancel each other out, thus yielding

$$\begin{aligned} c_k^{\text{CM}} &= -\frac{1}{T_\Omega} \sum_{n=1}^{Z-1} [x(n) - x(n-1)] \\ &\quad \times \Delta_n \text{sinc}(k f_\Omega \Delta_n) e^{-j\pi k f_\Omega \Delta_n} e^{-j2\pi k f_\Omega n T_g}. \end{aligned} \quad (\text{B.4})$$

Using the approximation

$$\text{sinc}(k f_\Omega \Delta_n) e^{-j\pi k f_\Omega \Delta_n} \approx 1 \quad (\text{B.5})$$

that holds whenever the accumulated jitter is much smaller than the period T_Ω , the final expression can be obtained, i.e.,

$$c_k^{\text{CM}} = -\frac{1}{T_\Omega} \sum_{n=1}^{Z-1} [x(n) - x(n-1)] \Delta_n e^{-j\frac{2\pi}{Z} k n}. \quad (\text{B.6})$$

APPENDIX C

In the presence of a clock modulation, the actual clock frequency $f_{\text{ck}}(t)$, oscillates around the nominal value f_g , at a frequency f_m , according to

$$f_{\text{ck}}(t) = f_g (1 + \Lambda_P \sin(2\pi f_m t)). \quad (\text{C.1})$$

The clock period $T_{\text{ck}}(t)$ can be expressed in turn as

$$\begin{aligned} T_{\text{ck}}(t) &= \frac{1}{f_{\text{ck}}(t)} = \frac{1}{f_g (1 + \Lambda_P \sin(2\pi f_m t))} \\ &\approx T_g (1 - \Lambda_P \sin(2\pi f_m t)). \end{aligned} \quad (\text{C.2})$$

The accumulated jitter $\Delta(t)$ can be derived as average value of the deviations produced by the oscillations on time intervals of duration T_g , namely

$$\begin{aligned}\Delta(t) &= -\frac{1}{T_g} \int_t^{t+T_g} \Lambda_P T_g \sin(2\pi f_m t) dt \\ &= \frac{\Lambda_P}{2\pi f_m} [\cos(2\pi f_m t + T_g) - \cos(2\pi f_m t)] \\ &= -\frac{\Lambda_P}{\pi f_m} \sin(\pi f_m T_g) \sin\left(2\pi f_m t + \frac{T_g}{2}\right). \quad (C.3)\end{aligned}$$

For the digital sequence that describes the accumulated jitter, disregarding phase information, the following expression can be used:

$$\Delta_n = \frac{\Lambda_P}{\pi V f_\Omega} \sin\left(\pi \frac{V}{Z}\right) \cos\left(2\pi \frac{V}{Z} n\right). \quad (C.4)$$

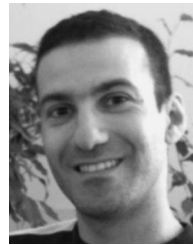
REFERENCES

- [1] *IEEE Standard for Terminology and Test Methods for Analog-to-Digital Converters*, IEEE Std. 1241-2000, Jun. 13, 2001.
- [2] E. Balestrieri, P. Daponte, and S. Rapuano, *Digital to Analog Converters: A Metrological Overview*. Piscataway, NJ: IEEE Press, 2005.
- [3] L. Angrisani, M. D'Apuzzo, and M. D'Arco, "Fast transformation for DAC parameters identification," *IEEE Trans. Instrum. Meas.*, vol. 55, no. 6, pp. 2006–2013, Dec. 2006.
- [4] B. Razavi, *Principles of Data Conversion System Design*. Piscataway, NJ: IEEE Press, 1995.
- [5] P. P. Fasang, "Optimal selection of test vectors for DA converter testing," *Measurement*, vol. 31, no. 3, pp. 165–174, Apr. 2002.
- [6] Y.-C. Jenq, "Direct digital synthesizer with jittered clock," *IEEE Trans. Instrum. Meas.*, vol. 46, no. 3, pp. 653–655, Jun. 1997.
- [7] N. Kurosawa, H. Kobayashi, H. Kogure, T. Komuro, and H. Sakayori, "Sampling clock jitter effects in digital-to-analog converters," *Measurement*, vol. 31, no. 3, pp. 187–199, Apr. 2002.
- [8] Y.-C. Jenq, "Digital-to-analog (D/A) converters with nonuniformly sampled signals," *IEEE Trans. Instrum. Meas.*, vol. 45, no. 1, pp. 56–59, Feb. 1996.
- [9] Y.-C. Jenq, "Digital spectra of non-uniformly sampled signals: Fundamentals and high-speed waveform digitizers," *IEEE Trans. Instrum. Meas.*, vol. 37, no. 2, pp. 245–251, Jun. 1988.
- [10] J. T. Doyle, Y. J. Lee, and Y.-B. Kim, "Fast and accurate DAC modeling techniques based on wavelet theory," *Microelectron. J.*, vol. 35, no. 5, pp. 451–460, May 2004.
- [11] M. Kosunen, J. Vankka, F. Teikari, and K. Halonen, "DNL and INL yield models for a current-steering D/A converter," in *Proc. ISCAS*, May 25–28, 2003, vol. 1, pp. 969–972.
- [12] R. Tortosa, J. M. de la Rosa, F. V. Fernández, and A. Rodríguez-Vázquez, "Clock jitter error in multi-bit continuous-time sigma-delta modulators with non-return-to-zero feedback waveform," *Microelectron. J.*, vol. 39, no. 1, pp. 137–151, Jan. 2008.
- [13] J. De Maeyer, P. Rombouts, and L. Weyten, "Addressing static and dynamic errors in unit element multibit DACs," *Electron. Lett.*, vol. 39, no. 14, pp. 1038–1039, Jul. 10, 2003.
- [14] *Agilent 33220 15 MHz Function/Arbitrary Waveform Generator User's Guide*, Agilent Technol., Santa Clara, CA, Dec. 1999, Publication number 33120-90014, Edition 6.
- [15] G. A. Friel, D. M. Hummels, and F. H. Irons, "Measurement and compensation of digital-to-analog converter nonlinearity," *Measurement*, vol. 31, no. 3, pp. 175–186, Apr. 2002.
- [16] L. Angrisani and M. D'Arco, "Modeling timing jitter effects in digital-to-analog converters," *IEEE Trans. Instrum. Meas.*, vol. 58, no. 2, pp. 330–336, Feb. 2009.
- [17] A. Baccigalupi and M. D'Arco, "An experimental study aimed at analysing horizontal quantization and time-base jitter effects in waveforms generated by means of DACs," in *Proc. 12th IMEKO TC-4 Workshop ADC Modeling Testing*, vol. 1, Iasi, Romania, Sep. 19–21, 2007.
- [18] M. D'Arco, A. Liccardo, and M. Vadursi, "Spectral estimation of piece-constant waveforms generated by digital-to-analog converters," in *Proc. IEEE I2MTC*, Singapore, May 5–7, 2009, pp. 1479–1485.



Massimo D'Apuzzo was born in Napoli, Italy, in 1947. He received the M.S. degree in chemical engineering and electrical engineering from the Università degli Studi di Napoli Federico II, Napoli, in 1971 and 1973, respectively.

In 1974, he joined the Dipartimento di Ingegneria Elettrica, the Università degli Studi di Napoli Federico II, as an Assistant Professor of electrical measurements and became an Associate Professor of electrical measurements in 1980. From 1986 to 1988, he was a Full Professor of electrical measurements with the University of L'Aquila, L'Aquila, Italy. From 1989 to 1991, he was a Full Professor of electrical measurements with the University of Salerno, Salerno, Italy. Since 1991 he has been a Full Professor of electrical measurements with the Università degli Studi di Napoli Federico II, where he is currently the President of the Division of Sciences and Technologies. His current research interests include new approaches for measurement uncertainty evaluation, the use of wavelet transform for measurement applications, and new methods based on time–frequency transforms for testing RF equipment for mobile communications.



Mauro D'Arco was born in Salerno, Italy, in 1974. He received the M.S. degree in electronic engineering from the University of Salerno, Salerno, in 1999 and the Ph.D. degree in electrical engineering from the Università degli Studi di Napoli Federico II, Napoli, Italy, in 2003.

He is currently an Assistant Professor with the Dipartimento di Ingegneria Elettrica, the Università degli Studi di Napoli Federico II. His research interests include digital signal processing techniques for instantaneous frequency estimation and RF equipment troubleshooting, new methods for phase-noise measurements, modeling and testing of A/D and D/A converters, and digital signal processing techniques for power quality measurements.



Annalisa Liccardo received the M.S. and Ph.D. degrees in electrical engineering from the Università di Napoli Federico II, Napoli, Italy, in 2003 and 2006, respectively.

She is with the Dipartimento di Ingegneria Elettrica, Università di Napoli Federico II, where she is currently a Researcher. Her current research interests include the implementation of distributed measurement systems, sensor networks for power quality measurements, and dynamic characterization of analog-to-digital and digital-to-analog converters.



Michele Vadursi (M'08) was born in Napoli, Italy, on May 9, 1978. He received the M.S. degree (with honors) in telecommunication engineering and the Ph.D. degree in electrical engineering from the Università di Napoli Federico II, Napoli, in 2002 and 2005, respectively.

Since 2006, he has been an Assistant Professor with the Dipartimento per le Tecnologie, Università degli Studi di Napoli Parthenope, Napoli. He has taken part in the European project FP7-216295 CA AMBER "Assessing, Measuring and BEenchmarking Resilience." His research interests include the design, implementation, and characterization of innovative measurement methods based on digital signal processing for testing communication systems and apparatuses, as well as communication network test and measurement and measurements on distributed systems and algorithms.

Dr. Vadursi was the recipient of the IET Communications Premium Award 2009 for the paper entitled "Performance measurement of IEEE 802.11b-based networks affected by narrowband interference through cross-layer measurements" (*IET Communications*, vol.2, No.1, Jan. 2008, pp.82–91).